Akoizo Original Practice - Form 2

Division B | Meteorology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) A station model pressure code of 984 means:

A. 984 mb

B. 1008.4 mb

C. 998.4 mb

D. 1098.4 mb

Answer: _____

2. (2 points) Which air mass is warm and humid?

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
Worked-case data: On a spring afternoon in Oklahoma, a dryline lies near the Texas border, mT air with dew points near 20 °C is to its east, a strong jet streak passes overhead, and winds turn from southeast at the surface to west aloft. Assess the severe weather threat.
Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. Continental polar

B. Continental tropical

C. Continental arctic

D. Maritime tropical

Answer: _____

3. (2 points) Damage spreading outward in a starburst pattern indicates:

Unit: Damaging winds and tornadoes.
Worked-case data: A damage survey finds trees in a field all blown down pointing outward from a central point, and a well-built house nearby lost its roof. Was this a tornado?
Separate investigation observations:
Damage path: A narrow path 300 m wide and 12 km long.
Debris: Trees fell in different directions, converging toward the path center.
Homes: Well-built homes lost exterior walls; interior rooms remained.

A. A downburst

B. A tornado

C. Hail

D. A dryline

Answer: _____

4. (2 points) Sleet forms when:

Unit: Precipitation, floods, winter storms, drought, and safety.
Worked-case data: A sounding shows below-freezing air from 3 km to the cloud top, a layer from 1.5 to 3 km at +3 °C, and a shallow layer from the ground to 300 m at -2 °C. What precipitation will reach the ground?
Separate investigation observations:
Rainfall: 150 mm fell in three hours over a small hilly basin.
Land surface: A wildfire burned the slopes last summer.
Stream gauge: Water rose 3 m in 90 minutes.

A. Hail falls from a supercell

B. Melted snow refreezes in a deep cold layer near the ground

C. Snow never melts

D. Rain freezes on contact

Answer: _____

5. (2 points) In a METAR, “31/22” means:

A. Pressure 31.22 inches

B. Temperature 31 °C, dew point 22 °C

C. Wind 310° at 22 knots

D. Visibility 31 miles

Answer: _____

6. (2 points) A dryline separates air masses that differ mainly in:

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Wind shift: Southwest winds west of the line, southeast winds east.

A. Pressure only

B. Wind speed only

C. Moisture

D. Temperature

Answer: _____

7. (2 points) The EF scale estimates wind speed from:

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Separate investigation observations:
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Debris: Trees fell in different directions, converging toward the path center.
Homes: Well-built homes lost exterior walls; interior rooms remained.

A. Direct anemometer readings in the tornado

B. Radar reflectivity

C. Funnel width

D. Observed damage to indicators

Answer: _____

8. (2 points) A tornado warning means:

Unit: Precipitation, floods, winter storms, drought, and safety.
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Separate investigation observations:
Rainfall: 150 mm fell in three hours over a small hilly basin.
Land surface: A wildfire burned the slopes last summer.
Stream gauge: Water rose 3 m in 90 minutes.

A. A tornado is occurring or imminent; take shelter now

B. Conditions may become favorable later

C. The storm has ended

D. Only hail is expected

Answer: _____

9. (2 points) A long hodograph that curves clockwise near the ground indicates:

Unit: Reading weather data: station models, METARs, hodographs, and models.
Worked-case data: A station model shows 88 at upper left, 70 at lower left, a wind barb from the south with two long barbs and one short barb, and 096 at upper right. Decode it.
Separate investigation observations:
METAR: KDDC 151900Z 20030G45KT 7SM TSRA BKN030CB 29/21 A2972
Wind: From 200° at 30 knots gusting to 45 knots.
Weather: Thunderstorm with rain and cumulonimbus clouds.
Pressure: Altimeter 29.72 inHg and falling quickly.

A. Strong shear favoring supercells

B. No wind

C. Stable air

D. Calm conditions favoring fog

Answer: _____

10. (2 points) Why do hurricanes weaken after landfall?

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A. Coriolis force becomes zero

B. They lose their warm-water energy source

C. Land air is too moist

D. The jet stream stops them

Answer: _____

11. (2 points) A small round area of high reflectivity at a hook's tip with low CC suggests:

A. Ground clutter only

B. A gust front

C. A debris ball from a tornado

D. Light snow

Answer: _____

12. (2 points) A "100-year flood" has what chance of occurring in any given year?

A. 100%

B. 10%

C. 0.1%

D. 1%

Answer: _____

13. (4 points) Explain how to read wind direction from a wind barb.

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Answer: _____
_____
_____

14. (4 points) Explain how wind shear helps a thunderstorm last longer.

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Answer: _____
_____
_____

15. (4 points) Why can two tornadoes with similar winds get different EF ratings?

Answer: _____
_____
_____

16. (4 points) Explain why ice crystals grow at the expense of supercooled droplets.

Answer: _____
_____
_____

17. (4 points) Decode “TSRA” and “+RA” in a METAR.

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18. (4 points) Describe how a lake-effect snow band forms.

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